TAN YONG JIAN

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CAREER SUMMARY

Engineering professional with 10 years of experience managing complex structural projects, coordinating technical matters across multiple stakeholders, and working with information-intensive operational processes. Brings hands-on experience in reporting automation, system monitoring, log analysis, and process improvement, supported by technical skills in Python, SQL, ETL/ELT pipelines, data transformation, database management, and cloud technologies. Currently transitioning into Data Engineering, with hands-on project experience building data pipelines and applying modern data engineering practices to support reliable, data-driven processes.

TECH SKILLS

- **Programming & Querying**: Python, PySpark, SQL, Shell Scripting
- **Data Engineering & Databases**: ETL/ELT Pipelines, Medallion Architecture, Data Warehousing, Star Schema, PostgreSQL
- **Cloud & DevOps**: Microsoft Fabric, Microsoft Azure, Docker, Git, GitHub Actions (CI/CD)
- **Data Visualisation**: Power BI
- Additional Languages: Java, JavaScript, C

PROFESSIONAL DEVELOPMENT

GENERATION SINGAPORE (IN PARTNERSHIP WITH MICROSOFT AND TEMASEK POLYTECHNIC)
JUNIOR DATA ENGINEER PROGRAM JUL 2026 – OCT 2026

- Completed a 12-week full-time training program developing core technical skills in collecting, storing, and retrieving data, building data pipelines using ELT principles, and deploying solutions on cloud platforms.
- Developed strong problem-solving capabilities, stakeholder communication, and collaboration practices tailored to modern tech environments.

Data Engineering Final Project | Olist Market Analysis

- Designed and implemented a **Medallion data pipeline** in **Microsoft Fabric**, ingesting nine Olist e-commerce datasets into **OneLake** and transforming raw data through Bronze, Silver, and Gold layers for analytics.
- Developed **PySpark** and **SQL** scripts for **automated** data cleaning, transformation, validation, and **data-quality checks**, and built a Gold-layer **star schema** with fact and dimension tables for business analysis and **Power BI** reporting.

Data Engineering Interim Project | Hospital Emergency Routing

- Designed and built a **multi-source ETL pipeline** in **Python** to extract live A&E waiting-time data via **Selenium web scraping**, and integrate dengue cluster and weather datasets from **REST APIs** (NEA, OneMap, data.gov.sg).
- Implemented automated data cleaning, deduplication, and postal-prefix standardization using **Pandas** and **SQL** to improve data quality and consistency.
- Developed **routing logic** to calculate "Total Time-To-Treatment" by combining hospital queue time, travel duration, and environmental risk factors, recommend a **data-riven** shortest duration hospital.

NANYANG TECHNOLOGICAL UNIVERSITY
(SCTP) Advance Professional Certificate in Software Engineering (Part-time) MAY 2024 – NOV 2024

- Completed an intensive software engineering program covering full-stack development, object-oriented principles, and the software development lifecycle (SDLC).
- Developed robust technical capabilities across frontend and backend technologies, including JavaScript, React, Java, Spring Boot, Hibernate, and PostgreSQL.

Software Engineering Capstone Project | Garang-Guni App | GoogleDrive_Link

- Designed and developed a full-stack recycling-material marketplace application as part of a 4-member team, implementing user authentication with JWT, collection scheduling, and relational database integration.
- Developed RESTful APIs, containerized services using Docker, and established automated CI/CD pipelines through GitHub Actions, incorporating testing and code-quality tools including Snyk, SpotBugs, JUnit, and Apache JMeter.
- Extended the application into a mobile application using React Native, implementing responsive components and structured navigation.

WORK EXPERIENCES

SOC L1 ANALYST
IRAS (via Spade Consulting) | Singapore OCT 2023 – NOV 2024

- Automated daily reporting and documentation workflows using Excel VBA, reducing manual-entry errors and reporting turnaround time.
- Monitored and processed system logs in a 24/7 SOC environment, escalating critical security threats according to SLA requirements.
- Reduced security response time by over 50% by fine-tuning SIEM detection use cases and alert logic.

PROJECT MANAGER & STRUCTURAL ENGINEER OCT 2014 – JUN 2023
GH Engineering, LongXiang Construction |HP Engineering, Tiong Aik Construction, Tat Hin Builders

- Managed structural delivery for large-scale HDB and JTC projects valued at S$3M–S$100M, leading teams of more than 100+ workers and coordinating technical matters across multidisciplinary teams and trades, including M&E, architecture, landscape, surveying, geotechnical, piling, and structural steel, while meeting technical, constructability, and safety requirements.
- Optimized resource allocation, data tracking, and project schedules to reduce inefficiencies, manage risks, and maintain delivery timelines.

CAREER BREAK DEC 2024 – JUL 2026

Took a planned career break for caregiving responsibilities while undertaking technical upskilling and preparing for a IT transition.

ADDITIONAL TECHNICAL TRAINING

42Singapore Piscine MAY 2023 – JUN 2023

- Completed an intensive one-month coding program covering algorithmic problem-solving, Unix/Linux environments, and C programming fundamentals.

Postgraduate Diploma in Cybersecurity (E-Learning) – PSB Academy MAR 2023 – OCT 2023

EDUCATION

- **Master of Construction Management (CGPA: 3.8/4.0)**
  Universiti Teknologi Malaysia (UTM), Malaysia AUG 2013 – OCT 2014
- **Bachelor of Science (Hons) in Construction Management (CGPA: 3.82/4.0)**
  Universiti Tunku Abdul Rahman (UTAR), Malaysia MAY 2010 – JUN 2013